

4	(a) (i) Displacement of particles perpendicular to direction of energy propagation	B1
	(ii) waves <u>meet</u> / overlap (at a point) (resultant) displacement is sum of the individual displacements	B1 B1
	(b) (i) $\lambda = vT$ or $\lambda = v/f$ and $f = 1/T$ $\lambda = 4.0 \times 1.5$ $\lambda = 6.0 \text{ (cm)}$	C1 A1
	(ii) path difference [= (44 cm – 29 cm)/6 cm] = 2.5λ	M1
	either waves have path difference = $(n + \frac{1}{2})\lambda$ or waves have phase difference = 180°	M1
	so destructive interference	A1
	(c) (i) intensity $\propto (\text{amplitude})^2$ ratio = $(0.60^2 / 0.90^2) = 0.44$	C1 A1
	(ii) phase difference = 90°	A1